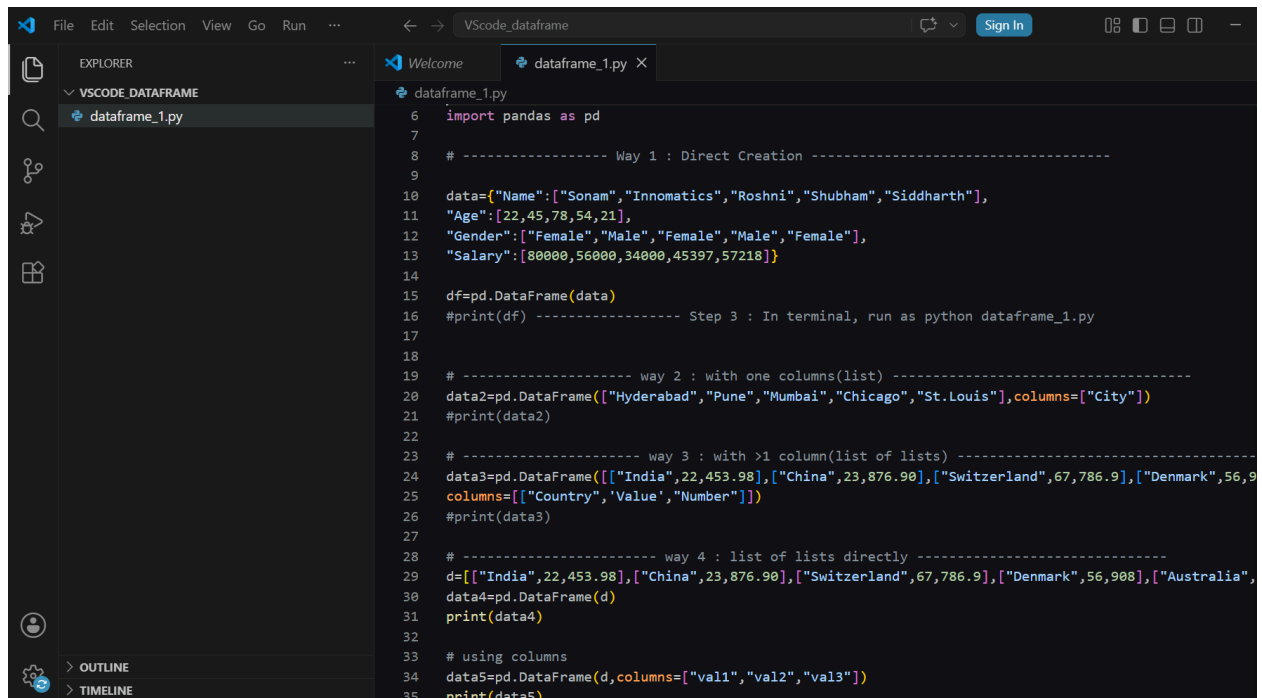
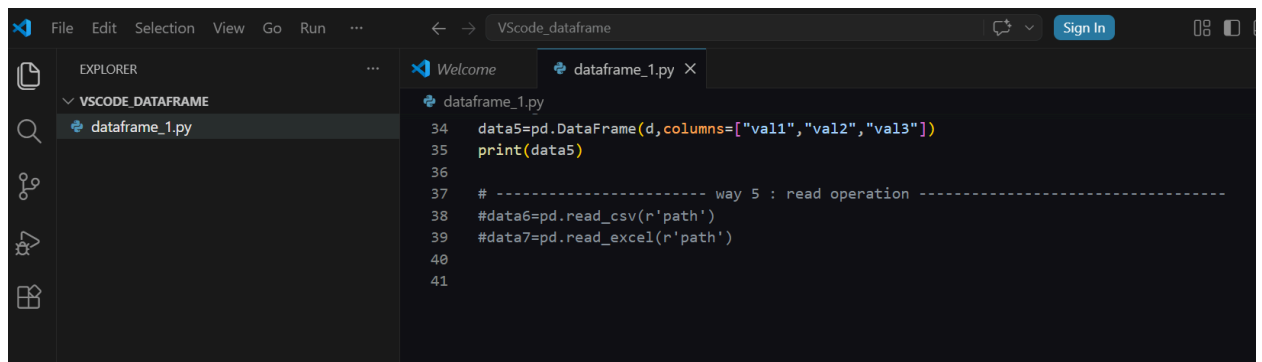


VS CODE Dataframe creation

Steps: without creating environment : New folder - new terminal(cmd prompt) - pip install pandas - file(dataframe_1.py) - logic code - run cmd as python filename(python dataframe_1.py)



```
6 import pandas as pd
7
8 # ----- Way 1 : Direct Creation -----
9
10 data={"Name":["Sonam","Innomatics","Roshni","Shubham","Siddharth"],
11       "Age":[22,45,78,54,21],
12       "Gender":["Female","Male","Female","Male","Female"],
13       "Salary":[80000,56000,34000,45397,57218]}
14
15 df=pd.DataFrame(data)
16 #print(df) ----- Step 3 : In terminal, run as python dataframe_1.py
17
18 # ----- way 2 : with one columns(list) -----
19 data2=pd.DataFrame(["Hyderabad","Pune","Mumbai","Chicago","St.Louis"],columns=["City"])
20 #print(data2)
21
22 # ----- way 3 : with >1 column(list of lists) -----
23 data3=pd.DataFrame([["India",22,453.98],["China",23,876.90],["Switzerland",67,786.9],["Denmark",56,908]],columns=["Country","Value","Number"])
24 #print(data3)
25
26 # ----- way 4 : list of lists directly -----
27
28 d=[["India",22,453.98],["China",23,876.90],["Switzerland",67,786.9],["Denmark",56,908],["Australia",23,456.78]]
29 data4=pd.DataFrame(d)
30 print(data4)
31
32 # using columns
33 data5=pd.DataFrame(d,columns=["val1","val2","val3"])
34 print(data5)
```



```
34 data5=pd.DataFrame(d,columns=["val1","val2","val3"])
35 print(data5)
36
37 # ----- way 5 : read operation -----
38 #data6=pd.read_csv(r'path')
39 #data7=pd.read_excel(r'path')
```

Step : Difference between creating dataframes with .py and .ipynb extensions ?

.py :

- Runs entire file at once

- Output appears in terminal
- Faster , lightweight, easier for version control

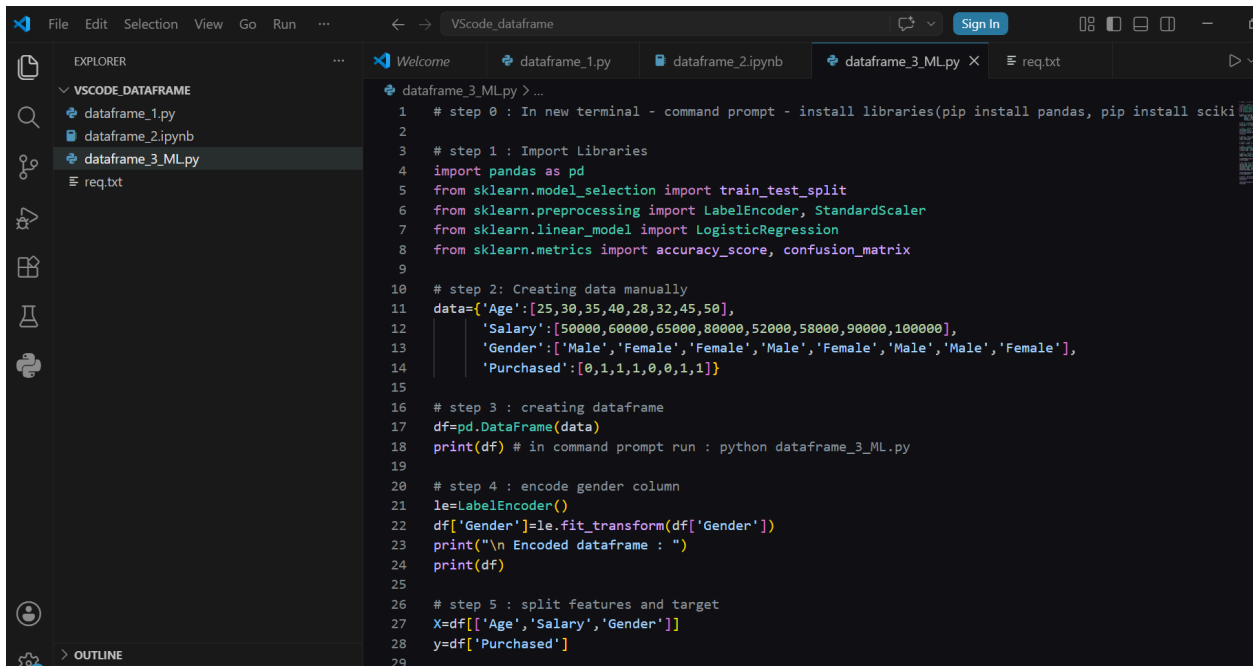
.ipynb :

- Runs cell by cell
- Like colab, output shown cell by cell
- Dataframes are created and shown automatically

Update : Dataframe creation for ML model

Step 1 : In terminal - pip install pandas, pip install scikit-learn

Step 2 : In new file - eg : dataframe_3-[ML.py](#) , write the code as below



```

1  # step 0 : In new terminal - command prompt - install libraries(pip install pandas, pip install scikit-learn)
2
3  # step 1 : Import Libraries
4  import pandas as pd
5  from sklearn.model_selection import train_test_split
6  from sklearn.preprocessing import LabelEncoder, StandardScaler
7  from sklearn.linear_model import LogisticRegression
8  from sklearn.metrics import accuracy_score, confusion_matrix
9
10 # step 2: Creating data manually
11 data={'Age':[25,30,35,40,28,32,45,50],
12       'Salary':[50000,60000,65000,80000,52000,58000,90000,100000],
13       'Gender':['Male','Female','Female','Male','Female','Male','Male','Female'],
14       'Purchased':[0,1,1,0,0,1,1]}
15
16 # step 3 : creating dataframe
17 df=pd.DataFrame(data)
18 print(df) # in command prompt run : python dataframe_3_ML.py
19
20 # step 4 : encode gender column
21 le=LabelEncoder()
22 df['Gender']=le.fit_transform(df['Gender'])
23 print("\n Encoded dataframe : ")
24 print(df)
25
26 # step 5 : split features and target
27 X=df[['Age','Salary','Gender']]
28 y=df['Purchased']
29

```

```
30 # step 6 : Train - test - split
31 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
32
33 # step 7 : feature scaling
34 scaler = StandardScaler()
35 X_train[['Age', 'Salary']] = scaler.fit_transform(X_train[['Age', 'Salary']]) # on train data to avoid da
36 X_test[['Age', 'Salary']] = scaler.transform(X_test[['Age', 'Salary']]) # on test data
37
38 # step 8 : logistic regression model
39 model = LogisticRegression()
40 model.fit(X_train, y_train)
41
42 y_pred = model.predict(X_test)
43 print(y_pred)
44
45 # new query for prediction
46 query = {'Age': [30], 'Salary': [70000], 'Gender': ['Female']}
47 new_data = pd.DataFrame(query)
48 new_data['Gender'] = le.transform(new_data['Gender'])
49 new_data[['Age', 'Salary']] = scaler.transform(new_data[['Age', 'Salary']])
50 prediction = model.predict(new_data)
51
52 print("New query prediction")
53 print(prediction)
54
55 # Step 9 : Evaluation
56 acc = accuracy_score(y_test, y_pred)
57 print(acc)
58
59 cm = confusion_matrix(y_test, y_pred)
```

To install all dependencies / libraries together , we can use **requirements file** :

- Add file as req.txt
- Place numpy, pandas, scikit-learn etc
- Pip install -r req.txt in command terminal and then proceed with code file as above , run it